

Annexure No.:

EXPERIMENT - 06

Aim: Interfacing Button-Controlled LED with Arduino board.

Apparatus: Arduino Board, wires, LED, Resistor, switch, Bread board.

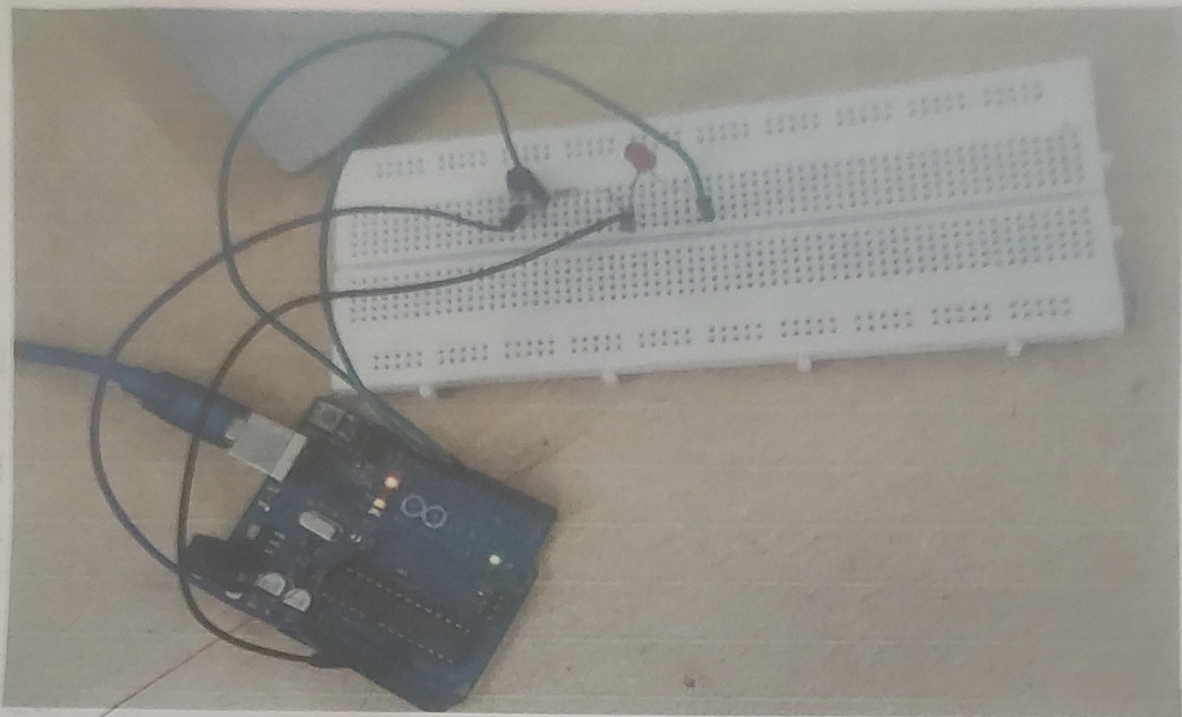
Theory:

Arduino circuit with an LED and a button.

- to build the circuit you will need those components:

- Arduino board (any board, if you don't have UNO you can easily adapt by finding +
- Breadboard.
- LED - any colour.
- Push button
- 220 ohm resistor for the LED. If you don't have this specific value, any resistor from 330 to 1k ohm will do.
- 10k ohm resistor for the push button. If you don't have, you can go until 20k-50k ohm.
- A bunch of male to male wires.

Performance result:



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CODE :

```
int s = 7;
int x;
int led = 8;

void setup() {
    serial.begin(9600);
    pinMode(7, INPUT);
    pinMode(8, OUTPUT);
}

void loop() {
    x = digitalRead(s);
    serial.println(x);
    if (x == 0)
    {
        digitalWrite(8, LOW);
    }
    else {
        digitalWrite(8, HIGH);
    }
}
```

Conclusion :

Shows button input controlling and LED using Arduino.

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10/4